Import Data Using Transform Maps (Spreadsheet)

Phase 1 - Brainstorming & Ideation

1. Problem Identification

Organizations often maintain large amounts of data in spreadsheets such as Excel or CSV files. Entering this data manually into ServiceNow can be time-consuming and may lead to data-entry errors, duplicate records, and inconsistent information.

The main problem identified is the need for an efficient way to import spreadsheet data into ServiceNow while ensuring that the data is correctly mapped, transformed, validated, and stored in the appropriate target table.

2. Brainstorming

During the brainstorming stage, several possible approaches were identified:

- Manually entering spreadsheet data into ServiceNow.
- Importing spreadsheet data using Import Sets.
- Using Transform Maps to map source fields to target fields.
- Using Transform Scripts for complex data transformation.
- Using Coalesce fields to prevent duplicate records.
- Validating the data before importing it into the target table.
- Implementing error handling for invalid or incomplete records.
- Automating regular imports using scheduled data imports.

Among these ideas, **Import Sets combined with Transform Maps** were identified as the primary approach for handling spreadsheet-based data imports.

3. Proposed Solution

The proposed solution is to import spreadsheet data into ServiceNow using an **Import Set and Transform Map**.

The overall process is:

Spreadsheet → Import Set → Import Set Table → Transform Map → Target Table

The spreadsheet acts as the source of data. The Import Set temporarily stores the imported data, while the Transform Map defines how the source fields should be mapped and transformed into the target ServiceNow table.

4. Field Mapping Idea

For example, consider a spreadsheet containing employee information:

Spreadsheet Field	ServiceNow Target Field
Employee ID	Employee ID
Employee Name	Name
Email	Email
Department	Department
Joining Date	Date of Joining

The Transform Map will establish the relationship between these source and target fields.

This ensures that each value from the spreadsheet is transferred to the correct field in the ServiceNow target table.

5. Data Transformation

The brainstorming process also considers situations where the spreadsheet data format does not match the target table format.

Possible transformations include:

- Converting date formats.
- Standardizing text values.
- Converting numbers into the required format.
- Removing unnecessary spaces or characters.
- Converting department names into appropriate reference values.
- Handling empty or missing values.
- Applying business-specific transformation rules.

For complex requirements, Transform Scripts can be considered to modify the data before it reaches the target table.

6. Duplicate Record Handling

Duplicate records are another important consideration.

A unique field such as **Employee ID** can be used as a **Coalesce field**.

The basic logic would be:

- If the Employee ID already exists → Update the existing record.
- If the Employee ID does not exist → Create a new record.

This helps maintain data consistency and reduces the possibility of duplicate records.

7. Validation and Error Handling

The solution should also consider data validation before and during transformation.

Examples include:

- Checking whether mandatory fields are populated.
- Validating email formats.
- Checking date formats.

- Identifying invalid department values.
- Detecting duplicate records.
- Recording transformation errors.
- Reviewing rejected or failed records.

This allows incorrect data to be identified and corrected instead of being blindly inserted into the target table.

8. Ideas Considered

Idea	Purpose
Manual Data Entry	Suitable only for small amounts of data
Direct Spreadsheet Import	Provides a simple way to load bulk data
Import Set	Temporarily stores imported source data
Transform Map	Maps and transforms source data into target fields
Transform Script	Handles complex transformation requirements
Coalesce	Helps identify existing records and avoid duplicates
Data Validation	Ensures data quality before insertion
Scheduled Import	Useful for recurring spreadsheet imports

9. Selected Idea

The selected approach is:

Spreadsheet + Import Set + Transform Map + Data Validation + Target Table

This approach provides a structured way to move bulk data from a spreadsheet into ServiceNow while maintaining proper field mapping, transformation, validation, and duplicate handling.

10. Expected Outcome

The expected outcomes of the proposed solution are:

- Reduce manual data-entry effort.
- Import large amounts of spreadsheet data efficiently.
- Map source fields accurately to target fields.
- Transform data into the required format.
- Reduce duplicate records.
- Identify invalid or incomplete data.
- Improve data consistency and accuracy.
- Simplify the overall data-import process.

11. Final Ideation Statement

The main idea of "Import Data Using Transform Maps (Spreadsheet)" is to provide an efficient and structured mechanism for importing spreadsheet data into ServiceNow. The solution uses Import Sets to load source data and Transform Maps to map, transform, validate, and transfer the data into the appropriate target table. This reduces manual effort, minimizes data-entry errors, supports duplicate handling, and improves overall data quality.